

# Hamiltonian Two-Way Coupling of Nonlinear Waves and 3D Flows

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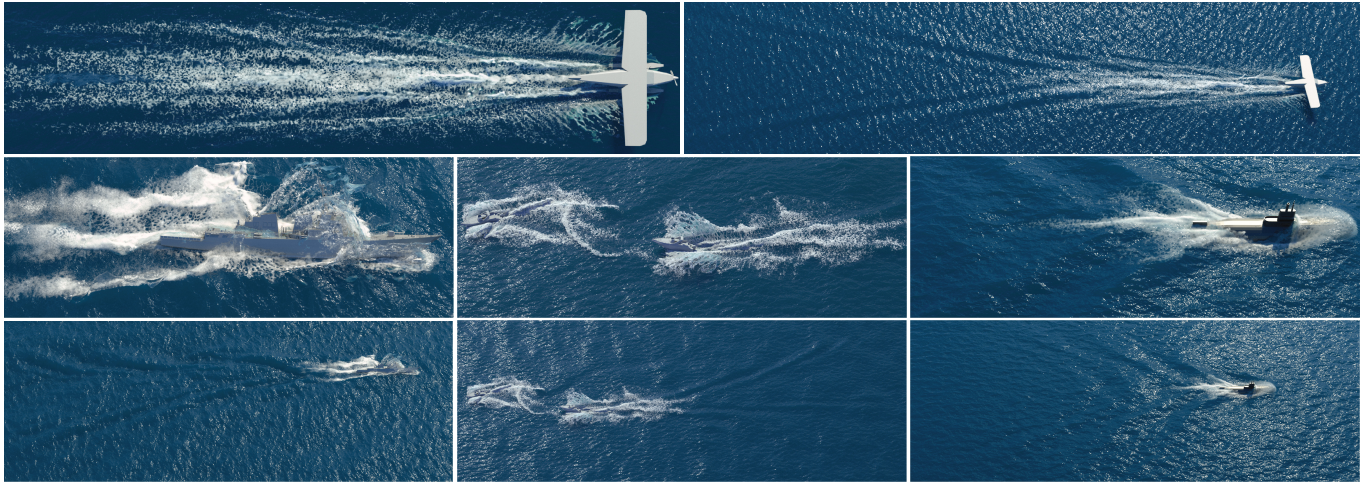


Fig. 1. A seaplane landing on the ocean surface, shown in near view (top-left) and far view (top-right). Near views of a battleship navigating through heavy waves (middle-left), two boats chasing each other (middle-center), and a submarine rising to the ocean surface (middle-right), with their corresponding far views shown in the bottom row.

Simulating large-scale free-surface flow by coupling a localized 3D fluid solver to a cheaper 2D surface model has long faced a mismatch in wave dynamics: efficient 2D wave models used in graphics are typically either linear or non-dispersive. These models are fast, simple, and accurate for calm, small-amplitude seas, but coupling them to strongly nonlinear 3D solvers produces visible reflections and artifacts at the 2D–3D interface. We address this problem by introducing a nonlinear and dispersive 2D wave model based on the canonical Zakharov formulation. Its Hamiltonian structure enables canonically consistent two-way coupling, allowing information to

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pass smoothly across the 2D–3D interface. Our 2D solver reduces mean wave-height error by factors of  $1.7\text{--}5\times$  relative to SWE, BEM, and Airy baselines while running more than  $10^3\times$  faster than BEM; it achieves greater nonlinear accuracy and coupling fidelity than SWE and Airy, with minor losses in speed and stability. Coupling it with a 3D Navier–Stokes solver yields a full system that suppresses visible seam artifacts across a range of experiments, including dispersion-matching and Kelvin-wake tests, and runs over  $4\times$  faster than a full-domain GPU NB-FLIP simulation.

CCS Concepts: • **Computing methodologies** → **Physical simulation**.

Additional Key Words and Phrases: Fluid simulation, wave simulation

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## 1 Introduction

Simulating large-scale free-surface flow remains a long-standing computational bottleneck in graphics, whether the scene is a seaplane splashing down on the open ocean, a battleship plowing through heavy seas, or a submarine breaking the ocean surface.

A fully three-dimensional incompressible Navier–Stokes (NS) simulation captures every regime of interest, including splashes, breaking waves, and fluid–solid coupling, but its memory and compute costs grow prohibitive once the domain spans hundreds of meters. A practical remedy is to confine the expensive 3D solver to a small embedding region and model the surrounding open ocean with a much cheaper 2D surface theory [Chentanez et al. 2015; Huang et al. 2021; Schreck and Wojtan 2022; Stomakhin and Selle 2017; Thürey et al. 2006] in order to propagate far-field waves efficiently. The success of this 2D–3D decomposition rests on the quality of the 2D model and the fidelity of the coupling between the two solvers.

Within this 2D–3D decomposition, a key obstacle to practical, high-fidelity coupling is that graphics currently lacks an efficient 2D wave model that is simultaneously *nonlinear and dispersive*: none retains the wave–wave interactions whose strength grows with wave steepness (nonlinearity) while allowing waves of different wavelengths to travel at their correct, wavelength-dependent speeds (dispersion). The two wave theories that dominate the literature both bypass the nonlinear surface dynamics. Airy wave theory [Airy 1845; Canabal et al. 2016; Schreck et al. 2019; Tessendorf 2014] linearizes the free-surface boundary conditions around the flat state, retaining the correct dispersion  $\omega^2 = gk \tanh(kh)$  but discarding every nonlinear interaction that gives rise to scenarios like the steepening of Stokes wavetrains. The discarded terms scale with wave steepness, so for the gentle ambient swell that Airy solvers typically animate, the linearization costs little and buys great speed and simplicity. The shallow-water equations (SWE) [Chentanez and Müller 2010; Kass and Miller 1990; Layton and van de Panne 2002] carry nonlinear depth-averaged advection but collapse the dispersion relation to  $\omega = \sqrt{gh} k$ , removing the very dispersive structure that distinguishes a real ocean from a sloshing tank, a sacrifice that is acceptable in the transport-dominated shallow flows SWE was built for, but not in open water. Lagrangian alternatives that have gained traction in graphics, e.g., wave particles [Yuksel et al. 2007], wave packets [Jeschke and Wojtan 2017], wavefront tracking [Jeschke and Wojtan 2015], and water-surface wavelets [Jeschke et al. 2018], all transport their primitives along linear ray paths and inherit the same linearization. Each of these trade-offs is defensible in its intended regime. Once strongly nonlinear waves dominate, however, these models lose accuracy, whereas our method makes the opposite trade-off, sacrificing a little speed and stability in exchange for greater nonlinear accuracy and coupling fidelity.

Coupling a localized 3D NS solver to such a linear 2D wave model creates precisely such a strongly nonlinear setting. The 3D solver is strongly nonlinear by construction, so waves reaching the embedding-box boundary exhibit nonlinear effects such as crest steepening and amplitude-dependent dispersion. A linear 2D model cannot represent any of these, so the two solvers disagree on the very content of the surface state at the interface. The disagreement shows up as spurious reflections at the seam and visible surface artifacts. We therefore identify the absence of a nonlinear and dispersive 2D wave model in graphics as a key obstacle to a 2D–3D coupling pipeline that is at once practical, accurate, and largely free of interface artifacts. The one prior nonlinear route, the boundary-element method (BEM) [Huang et al. 2021], recovers the nonlinear

free-surface potential flow at the PDE level, but discretizes it on a Lagrangian surface mesh, and pays for its nonlinear accuracy with per-step surface tracking/remeshing, dense  $O(N^3)$  boundary interactions, and high implementation complexity.

We address this limitation by introducing a 2D Eulerian wave model that is at once *nonlinear and dispersive*, and we show that its underlying *Hamiltonian* structure is what makes the two-way coupling to a 3D NS solver natural. The wave model evolves the canonical Zakharov pair [Zakharov 1968]: the surface elevation  $\eta$  and the surface value  $\psi$  of the velocity potential. Here *Hamiltonian* means that  $(\eta, \psi)$  form a canonical pair in the sense of classical mechanics: they evolve like position and momentum under the total wave energy, the Hamiltonian (Section 3). We advance the Zakharov system using the Craig–Sulem expansion of the Dirichlet–Neumann operator (DNO)  $G(\eta)$  [Craig and Sulem 1993]. This reduces every nonlinear correction to FFTs and pointwise products on a regular grid at  $O(N^2 \log N)$  cost. In practice, we retain this expansion through second or third order (Section 3). The same DNO machinery enables two-way coupling: it lifts  $(\eta, \psi)$  to depth-resolved boundary velocities for the 3D solver and, in reverse, reconstitutes a canonical wave state after relaxing  $\eta$  toward the 3D surface.

On standalone 2D wave-propagation benchmarks, our wave model alone lowers the mean wave-height error by  $1.7\times$  (deep water) to  $2.4\times$  (shallow water) over the strongest linear baseline (Airy with dispersion kernels) and by 3 to  $5\times$  over SWE and BEM, while running more than  $10^3\times$  faster than BEM; the price of this accuracy is a modestly more expensive and slightly less stable wave step than the linear baselines, a few milliseconds per substep in absolute terms (Table 3). Once coupled to the 3D NS solver, a dispersion-matching test in which a wave packet radiates from the 3D box across the seam and a single-boat experiment in which a hull radiates a Kelvin wake through the seam together show that our coupled pipeline yields symmetric wave fields with minimal seam artifacts, whereas prior coupling schemes exhibit varying degrees of asymmetric propagation, spurious reflection, or visible 2D–3D inconsistency (Section 8); on the single-boat scene, our solver also runs more than  $4\times$  faster than a GPU NB-FLIP simulation of the same domain, i.e., the bounding box of the entire visible water region. We further run a suite of large-scale scenarios, a battleship in heavy seas, a surfacing submarine, two chasing boats, and a seaplane landing on the open ocean, in which the 2D–3D interface remains largely free of visible artifacts.

*Contributions.* Our main contributions are:

- (1) **A nonlinear and dispersive Eulerian wave model for graphics.** We evolve the canonical Zakharov pair  $(\eta, \psi)$  under the exact water-wave Hamiltonian on a structured grid at FFT cost, capturing nonlinear surface phenomena such as the steepening of Stokes wavetrains. This retains the nonlinear modeling capability of the prior BEM approach while reducing computational complexity from  $O(N^3)$  to  $O(N^2 \log N)$  without requiring per-step surface tracking or remeshing.
- (2) **A canonically consistent two-way coupling between  $(\eta, \psi)$  and a 3D NS solver.** Both directions build on the DNO expansion: the forward direction harmonically extends  $(\eta, \psi)$  to depth-resolved bulk velocities on the embedding-box faces, while the

reverse direction relaxes the wave model's  $\eta$  toward the 3D free-surface elevation and reconstitutes  $\psi$  from the relaxed  $\eta$  and the wave's own surface velocity through an inverse DNO solve, keeping the wave state canonical at every substep.

- (3) **A continuously tunable nonlinear wave model.** A scalar perturbation parameter  $\varepsilon \in [0, 1]$  modulates every nonlinear correction in the truncated DNO and the Bernoulli equation for stability, parametrizing a continuous family of solvers that degenerates to linear Airy theory at  $\varepsilon = 0$  and recovers the nonlinear (truncated) Zakharov system at  $\varepsilon = 1$ .

## 2 Related Work

### 2.1 2D simulation

**SWE.** The SWE, a simplified form of the Navier–Stokes equations, originated in engineering physics [Crapper 1984; Le Méhauté 2013; Stoker 2019] and were first introduced to computer graphics by Kass and Miller [1990]. Layton and van de Panne [2002] introduced an implicit semi-Lagrangian scheme to improve stability and support larger time steps. Chentanez and Müller [2010] extended SWE to handle arbitrary slopes and depths. Later works enhanced SWE by adding surface tension [Wang et al. 2007], reducing numerical dissipation while ensuring stability via implicit Newmark integration [Angst et al. 2008], and improving efficiency using a collocated grid and a simplified SWE formulation [Lee and O’Sullivan 2007].

**Airy Wave Theory.** Airy wave theory [Airy 1845; Birkhoff 2015; Dean and Dalrymple 1991], popularized in computer graphics by Tessendorf [2001], synthesizes ocean height fields by evolving wave spectra according to dispersion relations. Some implementations augment linearly evolved waves with geometric displacement or post-sharpening [Stomakhin and Selle 2017; Tessendorf 2001]; we distinguish this geometric nonlinearity from the dynamical nonlinearity in our wave evolution. Horvath [2015] further improved spectral realism using an advanced empirical model, and proposed an artist-friendly parameterization scheme to facilitate intuitive control over ocean wave elongation. Beyond Tessendorf’s formulation, Stam [2001] proposed a compact FFT-based fluid solver widely used in 2D surface simulations. Nielsen et al. [2013] fit artist-created wave fields to spectral parameters for consistent high-frequency synthesis. Later, Tessendorf [2014] presented the *eWave* model, which provides a time integration in both Fourier and convolution forms, enabling more flexible boundary treatments. Dispersion-kernel approaches [Canabal et al. 2016] handle solid boundaries via convolution and shadow masks. More recently, Schreck et al. [2019] introduced a fundamental solution-based method using Green’s functions to capture wave–obstacle interactions in open domains without grids.

**High-Order Spectral Method.** Starting from the canonical Hamiltonian formulation of Zakharov [1968], the high-order spectral (HOS) method was independently introduced by Dommermuth and Yue [1987] and West et al. [1987] to evolve nonlinear surface waves on a flat reference grid via a Taylor expansion of the Dirichlet–Neumann operator, with Craig and Sulem [1993] casting the expansion in the explicit operator form used in modern HOS solvers. Such interior-to-boundary reductions via Dirichlet–Neumann (Steklov–Poincaré) operators also appear in graphics, including extrinsic

shape analysis [Wang et al. 2018], boundary-first conformal flattening [Sawhney and Crane 2017], and boundary-only physics-based skinning [Gao et al. 2014]. Subsequent work extended HOS to directional sea states [Bateman et al. 2001], variable bathymetry [Guyenne and Nicholls 2008], and large-scale open-ocean simulation [Ducroz et al. 2016]. We adopt the HOS expansion as the engine of our 2D wave solver.

**Other Wave Simulation.** A variety of approaches have explored alternative physical representations to better capture dispersion, boundary interactions, and visual detail. These include wave particles [Yuksel et al. 2007], wave packets [Jeschke and Wojtan 2017], wavefront tracking [Jeschke and Wojtan 2015], and water surface wavelets [Jeschke et al. 2018]. Hybrid models combine different regimes, such as SWE with Airy theory [Jeschke and Wojtan 2023], or augment height fields with particles for breaking waves [Thürey et al. 2007]. Others enhance detail as a post-process, like wave curves on deforming surfaces [Skrivan et al. 2020].

### 2.2 3D simulation

Early works in 3D fluid simulation for graphics [Foster and Fedkiw 2001; Foster and Metaxas 1996; Stam 1999] primarily adopt Eulerian frameworks, but often suffer from numerical dissipation. Pure Lagrangian approaches, such as smoothed particle hydrodynamics (SPH) [Ihmsen et al. 2014; Müller et al. 2003], require costly neighbor searches and struggle to enforce incompressibility. Hybrid Lagrangian–Eulerian methods such as Particle-in-Cell (PIC) [Harlow 1964] alleviate some issues but still exhibit dissipation. To address this, the Fluid-Implicit Particle (FLIP) method was introduced to graphics by Zhu and Bridson [2005], building on Brackbill et al. [1988], significantly reducing dissipation. To improve efficiency, Narrow Band FLIP (NB-FLIP) [Ferstl et al. 2016] and extended NB-FLIP [Sato et al. 2018] were later proposed. Furthermore, several approaches [Ando and Batty 2020; Losasso et al. 2006, 2004; Nielsen and Bridson 2016; Wang et al. 2025a; Zhu et al. 2013] have employed adaptive techniques to resolve fine-scale fluid features while maintaining computational efficiency for large-scale scenes. Recently, flow-map-based methods [Chen et al. 2025a, 2024, 2025b; Deng et al. 2023; He et al. 2026; Li et al. 2024, 2026, 2025a,b; Nabizadeh et al. 2022; Sun et al. 2024, 2025; Wang et al. 2026, 2024, 2025b; Zhou et al. 2024] and the Coadjoint Orbit FLIP [Nabizadeh et al. 2024] have reduced numerical dissipation and better preserved vortices.

### 2.3 Coupled 2D and 3D simulation

Thürey et al. [2006] coupled shallow water with a 3D solver using the Lattice Boltzmann method. Nielsen and Bridson [2011] improved art-directability by introducing guide shapes to constrain high-resolution liquid simulations to low-resolution inputs. Chentanez et al. [2015] combined SWE, SPH, and a 3D Eulerian solver, enabling efficient simulation but lacking accurate wave dispersion. Huang et al. [2021] addressed this by coupling BEM [Da et al. 2016; Huang and Michels 2020] with FLIP for more realistic ocean–ship interaction, though BEM’s mesh operations are computationally expensive and limit scalability. The fluxed animated boundary (FAB) method [Stomakhin and Selle 2017] allows artistic control but only supports one-way coupling (the 3D simulation cannot affect the

Table 1. Comparison with prior 2D–3D coupled water-simulation methods. “GPU-native” means that the 2D solver advances a height field stored on a uniform structured grid, making it well suited for GPU implementation; “Mesh-op-free” means the 2D solver requires no per-step mesh operations on an unstructured surface mesh (mesh construction, tracking, or remeshing) during simulation, in contrast to BEM-style methods that maintain a Lagrangian surface mesh; “Warmup-free” means the 2D solver is usable from  $t = 0$  without first accumulating a temporal sampling window; “Non-linear” means nonlinear wave dynamics rather than geometric sharpening; “Dispersive” means the 2D wave model reproduces the correct deep-water dispersion relation, so waves of different wavelengths propagate at different speeds; “Hamiltonian” means the 2D solver evolves the canonical Zakharov pair  $(\eta, \psi)$  under the water-wave Hamiltonian (Section 3).

| Method                         | Hamiltonian | Two-way | Nonlinear | GPU-native | Mesh-op-free | Warmup-free | Dispersive |
|--------------------------------|-------------|---------|-----------|------------|--------------|-------------|------------|
| Thürey et al. [2006]           | ✗           | ✓       | ✓         | ✓          | ✓            | ✓           | ✗          |
| FAB [Stomakhin and Selle 2017] | ✗           | ✗       | ✗         | ✓          | ✓            | ✓           | ✗          |
| Chentanez et al. [2015]        | ✗           | ✓       | ✓         | ✓          | ✓            | ✓           | ✗          |
| Schreck and Wojtan [2022]      | ✗           | ✓       | ✗         | ✓          | ✓            | ✗           | ✓          |
| Huang et al. [2021]            | ✗           | ✓       | ✓         | ✗          | ✗            | ✓           | ✓          |
| Ours                           | ✓           | ✓       | ✓         | ✓          | ✓            | ✓           | ✓          |

2D simulation). Perfectly Matched Layers (PML)-based methods [Berenger 1994; Bojsen-Hansen and Wojtan 2016; Söderström et al. 2010] reduce reflection at the interface via non-reflecting boundaries and allow localized re-simulation, but are also one-way. More recently, Schreck and Wojtan [2022] coupled a 2D fundamental solution solver [Schreck et al. 2019] with a 3D FLIP simulation, using dispersion matching to reduce coupling artifacts. Nonetheless, its underlying wave model remains linear Airy theory and therefore does not capture nonlinear wave effects. Table 1 summarizes how our method compares to representative prior coupled 2D–3D water solvers. To our knowledge, ours is the first to combine nonlinear and dispersive surface-wave dynamics with two-way coupling and ocean-scale efficiency in a single framework.

### 3 Hamiltonian Spectral Wave Solver

We model the open-ocean portion of the domain as the free surface of an incompressible, irrotational, inviscid fluid of finite depth  $-d \leq y \leq \eta(x, z, t)$  under gravity  $g$ . Throughout this section we let  $\mathbf{x} = (x, z)$  denote the horizontal coordinates,  $y$  the vertical coordinate,  $\eta(\mathbf{x}, t)$  the surface elevation,  $\Phi(\mathbf{x}, y, t)$  the bulk velocity potential ( $\mathbf{u} = \nabla\Phi$ ,  $\Delta\Phi = 0$ ), and  $\psi(\mathbf{x}, t) = \Phi(\mathbf{x}, \eta(\mathbf{x}, t), t)$  the velocity potential evaluated on the moving free surface. Our full method couples the 2D wave solver developed in this section to a localized 3D NS solver (Sections 4 and 5); to keep the two components clearly distinguished, we refer to the former as the *wave* model and to the latter as the *NS* solver; when we speak of the “2D (wave) side” and the “3D (NS) side”, we mean these two subdomains and their respective solvers. The dimension labels describe our default configuration; the same construction applies verbatim one dimension lower, where a 1D wave model embeds a 2D NS solver, as in some of our validation experiments (Section 8). Symbol conventions are summarized in Table 2.

#### 3.1 Canonical Zakharov formulation

Following Zakharov [1968], the water-wave problem admits a Hamiltonian formulation in which the canonical pair is exactly  $(\eta, \psi)$ . The

Table 2. Summary of the main symbols and notations.

| Notation             | Type     | Definition                                          |
|----------------------|----------|-----------------------------------------------------|
| $\eta$               | scalar   | free-surface elevation                              |
| $\psi$               | scalar   | surface velocity potential $\Phi _{\text{surface}}$ |
| $\Phi$               | scalar   | bulk velocity potential, $\mathbf{u} = \nabla\Phi$  |
| $G(\eta)$            | operator | Dirichlet–Neumann operator (DNO)                    |
| $G_0$                | operator | DNO at the flat state $\eta \equiv 0$               |
| $B_0$                | operator | Fourier multiplier with symbol $ \mathbf{k} ^2$     |
| $\mathbf{k}$         | vector   | horizontal wave vector                              |
| $\omega$             | scalar   | wave angular frequency, $\omega^2 = g G_0$          |
| $\mathbf{u}$         | vector   | velocity, components $(u, v, w)$                    |
| $\varepsilon$        | scalar   | nonlinear-amplitude parameter                       |
| $d$                  | scalar   | still-water depth                                   |
| $\sigma(\mathbf{x})$ | scalar   | relaxation source rate                              |
| $\phi$               | scalar   | 3D level-set field                                  |
| $t$                  | scalar   | time                                                |

fluid Hamiltonian is the sum of kinetic and potential energy, expressed entirely in surface variables,

$$H[\eta, \psi] = \frac{1}{2} \int \psi G(\eta) \psi d\mathbf{x} + \frac{1}{2} g \int \eta^2 d\mathbf{x}, \quad (1)$$

and the equations of motion are

$$\partial_t \eta = \frac{\delta H}{\delta \psi}, \quad \partial_t \psi = -\frac{\delta H}{\delta \eta}. \quad (2)$$

The single nontrivial object in this formulation is the Dirichlet–Neumann operator (DNO)  $G(\eta)$ , defined by

$$G(\eta)\psi(\mathbf{x}) = \sqrt{1 + |\nabla\eta|^2} \partial_n \Phi|_{y=\eta}, \quad (3)$$

i.e., the operator that, for each prescribed Dirichlet datum  $\psi$  on the moving surface, returns the normal derivative of the harmonic extension  $\Phi$  into the fluid. Knowing  $G(\eta)$  collapses the original three-dimensional Laplace problem to two surface fields  $(\eta, \psi)$  on a flat horizontal grid.

Carrying out the variational derivatives in (2) yields the canonical Zakharov system (Zakharov [1968], recast by Craig and Sulem [1993]),

$$\partial_t \eta = G(\eta) \psi, \quad (4)$$

$$\partial_t \psi = -g\eta - \frac{1}{2} |\nabla\psi|^2 + \frac{(G(\eta)\psi + \nabla\eta \cdot \nabla\psi)^2}{2(1 + |\nabla\eta|^2)}. \quad (5)$$

Equation (4) is the kinematic free-surface boundary condition expressed through the DNO; (5) is the exact Bernoulli equation for the surface potential. The system is closed: a numerical method that can evaluate  $G(\eta)\psi$  for arbitrary  $(\eta, \psi)$  on a regular grid yields a nonlinear ocean solver. Throughout the paper, “Hamiltonian” and “canonical” refer to this Zakharov state  $(\eta, \psi)$  and its energy structure. This structure is not a prerequisite for high-quality coupling in general; its value here is that  $(\eta, \psi)$  completely describes the irrotational surface state, so both directions of the 2D–3D exchange reduce to well-defined maps on these two fields: the 2D-to-3D direction (Section 4) lifts  $(\eta, \psi)$  to a depth-resolved bulk velocity, and

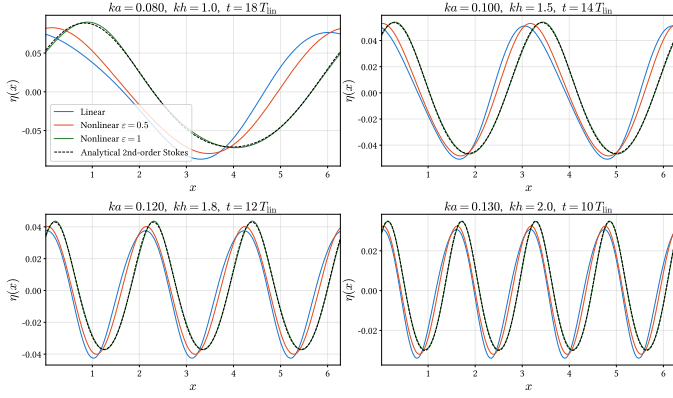


Fig. 2. Final-time surface elevation  $\eta(x)$  of the linear wave solver ( $\varepsilon = 0$ , blue, equivalent to [Canabal et al. 2016]), our nonlinear High-Order Spectral (HOS)-3 solver with  $\varepsilon = 0.5$  (red) and  $\varepsilon = 1$  (green), and the analytical second-order Stokes wave (black dashed) in four mild Stokes regimes, i.e., weakly nonlinear, non-breaking Stokes waves. The linear solver drifts out of phase due to the missing amplitude-dependent frequency shift, while the fully nonlinear ( $\varepsilon = 1$ ) HOS-3 solver stays phase-locked to the analytical wave. The  $\varepsilon = 0.5$  curve lies between the two.

the 3D-to-2D direction (Section 5) folds the 3D surface state back into a valid pair.

### 3.2 Spectral expansion of the Dirichlet–Neumann operator

A direct evaluation of  $G(\eta)\psi$  requires solving Laplace’s equation in the irregular fluid domain at every step. Craig and Sulem [1993] observed that, for  $\eta$  small relative to the characteristic wavelength  $1/k$ ,  $G(\eta)$  admits a convergent Taylor expansion in homogeneous powers of  $\eta$  that involves no Laplace solve at all. We adopt this expansion, known in ocean engineering as the High-Order Spectral (HOS) method [Dommermuth and Yue 1987], as the engine of our wave solver. Let  $G_0$  denote the operator at the flat reference state  $\eta \equiv 0$ . Its symbol on a horizontal Fourier mode  $e^{i\mathbf{k} \cdot \mathbf{x}}$  is

$$G_0(\mathbf{k}) = |\mathbf{k}| \tanh(|\mathbf{k}|d), \quad (6)$$

which is exactly the linear water-wave dispersion operator and reproduces  $\omega^2(\mathbf{k}) = g G_0(\mathbf{k})$  at leading order [Airy 1845]. Writing  $G(\eta) = \sum_{j \geq 0} G_j(\eta)$  with  $G_j$  homogeneous of degree  $j$  in  $\eta$ , the recursion of Craig and Sulem [1993] gives, in spectral form,

$$G_1(\eta)\psi = -G_0(\eta G_0\psi) - \nabla \cdot (\eta \nabla \psi), \quad (7)$$

$$G_2(\eta)\psi = G_0(\eta G_0(\eta G_0\psi)) - \frac{1}{2} G_0(\eta^2 B_0\psi) - \eta B_0(\eta G_0\psi) + \frac{1}{2} \eta^2 G_0(B_0\psi) + |\nabla \eta|^2 G_0\psi, \quad (8)$$

where  $B_0$  is the homogeneous Fourier multiplier with symbol  $|\mathbf{k}|^2$  (the second vertical derivative of the harmonic extension at the flat state). All operators in (7)–(8) are concatenations of Fourier multipliers ( $G_0, B_0, \nabla$ ) with pointwise products of  $\eta, \nabla \eta, \nabla \psi$ , and are therefore evaluated with FFTs and physical-space multiplications.

To improve stability in coupled simulations with steep 3D-driven surface features, we modulate every nonlinear term by a scalar

amplitude parameter  $\varepsilon \in [0, 1]$ ,

$$G(\eta) \approx G_0 + \varepsilon G_1(\eta) + \varepsilon^2 G_2(\eta), \quad (9)$$

and read  $\varepsilon = 1$  as the fully nonlinear model and  $\varepsilon = 0$  as the linear Airy limit. The dynamic equation (5) is rewritten consistently as

$$\partial_t \psi = -g\eta + \varepsilon \left[ -\frac{1}{2} |\nabla \psi|^2 + \frac{1}{2} (1 + \varepsilon^2 |\nabla \eta|^2) W^2 \right], \quad (10)$$

with

$$W = \frac{G_0\psi + \varepsilon G_1(\eta)\psi + \varepsilon^2 G_2(\eta)\psi + \varepsilon \nabla \eta \cdot \nabla \psi}{1 + \varepsilon^2 |\nabla \eta|^2}. \quad (11)$$

Throughout the paper we refer to the truncation in (9), which retains the three terms  $G_0, \varepsilon G_1, \varepsilon^2 G_2$ , as the Order-3 (or HOS-3) scheme; the Order-2 (HOS-2) variant referenced in our experiments retains only  $G_0$  and  $\varepsilon G_1$ .

We verify the effectiveness of this  $\varepsilon$  modulation in Section 8: the wave-propagation sweep against a high-resolution 3D Eulerian reference (Figure 14) and the Stokes-wave sweep against the analytical second-order Stokes solution (Figure 15) both show the height error decreasing monotonically as  $\varepsilon$  grows from 0 to 1, with the drop between  $\varepsilon = 0$  and  $\varepsilon \approx 0.1$  already accounting for a large fraction of the linear-model error. In other words, even a small  $\varepsilon$  already yields a significant improvement over the linear case. Conversely, in scenarios where strong forcing pushes the surface close to the breaking threshold and the higher-order DNO terms can excite numerical instabilities, we deliberately keep  $\varepsilon$  small to preserve solver stability:  $\varepsilon = 0.3$  for the submarine,  $\varepsilon = 0.2$  for the battleship in heavy waves, and  $\varepsilon = 0.2$  for the pond.

### 3.3 Boundary conditions via even extension

The FFT machinery above is intrinsically periodic; closed boundaries (a tank wall, a coastline, the perimeter of a pond) require additional care. We support reflecting and replicating boundaries through an even extension of the wave domain. Given the physical state  $(\eta, \psi)$  on a rectangle of resolution  $(M, N)$ , we form

$$\tilde{\eta}(\mathbf{x}) = \eta(R\mathbf{x}), \quad \tilde{\psi}(\mathbf{x}) = \psi(R\mathbf{x}), \quad (12)$$

on a doubled  $(2M, 2N)$  grid, where  $R$  denotes the appropriate reflection across each boundary. The operators  $G_0, G_1$ , and  $G_2$  are then constructed on the doubled grid and applied to  $\tilde{\eta}, \tilde{\psi}$ ; after the time step we crop the result back to  $(M, N)$ . The construction makes the discrete normal derivatives at the boundary vanish identically (Neumann condition) for reflecting walls, while a replicate variant approximates outflow. Because the extension only doubles the grid, the FFT cost increases only by a constant factor of four, and the boundary treatment carries no asymptotic overhead.

### 3.4 Time integration: exact-linear Adams–Bashforth

Following Craig and Sulem [1993], we split the system into an exactly integrated linear part and an AB2-advanced nonlinear part:

$$\partial_t \begin{pmatrix} \eta \\ \psi \end{pmatrix} = \mathcal{L} \begin{pmatrix} \eta \\ \psi \end{pmatrix} + \mathcal{N}(\eta, \psi), \quad (13)$$

where the linear part  $\mathcal{L} = \begin{pmatrix} 0 & G_0 \\ -g & 0 \end{pmatrix}$  is diagonal in Fourier space with eigenfrequency  $\omega(\mathbf{k}) = \sqrt{g G_0(\mathbf{k})}$ , and the nonlinear forcing  $\mathcal{N}$  collects all terms in  $\varepsilon$ . The linear part is stiff at high wavenumber,

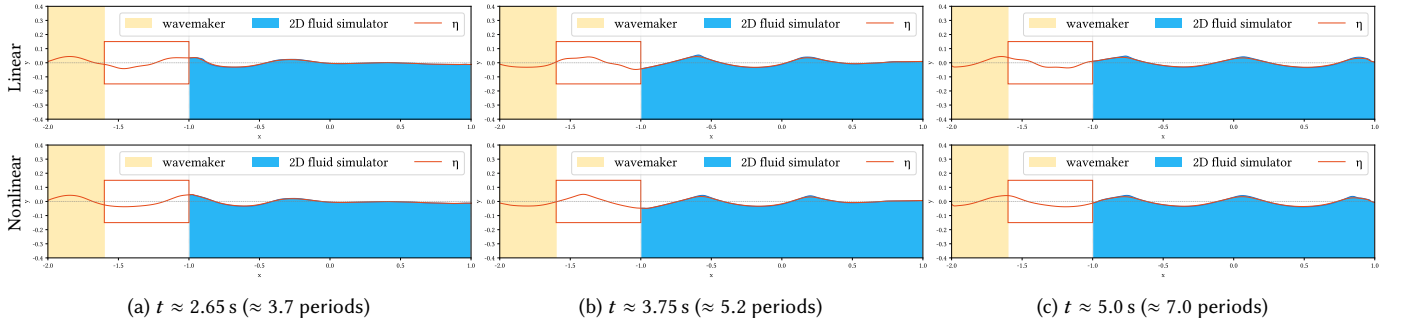


Fig. 3. Continuous Stokes wavetrain crossing the 2D-3D coupling interface, comparing the linear wave model ( $\varepsilon = 0$ , top row of each column) with our fully nonlinear HOS model ( $\varepsilon = 1$ , bottom row). A 1D wave domain ( $x \in [-2, 1]$  m, still-water depth 0.4 m) drives a second-order Stokes train of steepness  $ka = 0.3$  and wavelength 0.8 m through an embedded 2D fluid box ( $x \in [-1, 1]$  m, light blue fill); yellow shading marks the left-edge wavemaker zone. The red boxes outline the wave free-propagation region just outside the fluid box, where the 1D wave solver carries the train on its own. Inside the red boxes the linear surface develops visible kinks, is no longer smooth, and shows no crest–trough asymmetry, while the nonlinear surface remains smooth, exhibits the sharpened crests and flattened troughs of a Stokes train, and continues the surface shape inside the 2D fluid box almost exactly. The mismatch in the linear case reflects the incompatibility between the Airy wave model and the natively nonlinear 2D NS solver across the coupling interface at this steepness ( $ka = 0.3$ ).

i.e.,  $\omega(\mathbf{k})$  scales like  $\sqrt{|\mathbf{k}|}$  in deep water, so an explicit step on the full RHS would require  $\Delta t \lesssim 1/\omega(k_{\text{Nyq}})$ . We avoid the stiffness by integrating  $\mathcal{L}$  exactly. Per Fourier mode, the linear flow is the rotation

$$\begin{pmatrix} \hat{\eta} \\ \hat{\psi} \end{pmatrix}_{\text{lin}}(t + \Delta t) = \begin{pmatrix} \cos(\omega\Delta t) & \frac{G_0}{\omega} \sin(\omega\Delta t) \\ -\frac{g}{\omega} \sin(\omega\Delta t) & \cos(\omega\Delta t) \end{pmatrix} \begin{pmatrix} \hat{\eta} \\ \hat{\psi} \end{pmatrix}(t), \quad (14)$$

with the limit  $\hat{\eta} \rightarrow \hat{\eta}, \hat{\psi} \rightarrow \hat{\psi} - g\Delta t \hat{\eta}$  at  $\mathbf{k} = \mathbf{0}$ . The nonlinear forcing  $\hat{\mathcal{N}}_n = (\hat{F}_n^\eta, \hat{F}_n^\psi)$  is then advanced through the Duhamel response of (14), i.e., the variation-of-constants formula, which propagates the accumulated nonlinear forcing with the exact linear flow: setting  $\hat{F}_n^{q,\star} = \frac{3}{2}\hat{F}_n^q - \frac{1}{2}\hat{F}_{n-1}^q$ ,  $q \in \{\eta, \psi\}$  (an Adams–Bashforth-2 extrapolation), we update

$$\begin{aligned} \hat{\eta}_{n+1} &= \cos(\omega\Delta t) \hat{\eta}_n + \frac{G_0}{\omega} \sin(\omega\Delta t) \hat{\psi}_n \\ &\quad + \frac{\sin(\omega\Delta t)}{\omega} \hat{F}_n^{\eta,\star} + \frac{1 - \cos(\omega\Delta t)}{g} \hat{F}_n^{\psi,\star}, \end{aligned} \quad (15)$$

$$\begin{aligned} \hat{\psi}_{n+1} &= -\frac{g}{\omega} \sin(\omega\Delta t) \hat{\eta}_n + \cos(\omega\Delta t) \hat{\psi}_n \\ &\quad - \frac{1 - \cos(\omega\Delta t)}{G_0} \hat{F}_n^{\eta,\star} + \frac{\sin(\omega\Delta t)}{\omega} \hat{F}_n^{\psi,\star}. \end{aligned} \quad (16)$$

The first step (and any step in which  $\Delta t$  changes) bootstraps with  $F_0^\star = N_0$ , which reduces (15)–(16) to a first-order integrating-factor Euler step. The  $k = 0$  singularities cancel analytically:  $\sin(\omega\Delta t)/\omega \rightarrow \Delta t$  and  $(1 - \cos(\omega\Delta t))/G_0 \rightarrow \frac{1}{2}g\Delta t^2$ .

The scheme has three concrete advantages for our coupled setting. First, (14) integrates the linear dispersion exactly per Fourier mode, so the surface step is non-dissipative on linear waves regardless of  $\Delta t$ . Second, the AB2 extrapolation only touches the nonlinear forcing, which is bounded; the time-step constraint reduces from  $\omega(k_{\text{Nyq}})\Delta t < 1$  to a much milder CFL on  $N$ . Third, the same exponential-integrator strategy extends to the implicit relaxation source we add in Section 5: that source is linear in  $\eta - \eta^{\text{target}}$  with a position-dependent rate  $\sigma(x)$ , so it admits its own closed-form exponential decay (34), which we compose with the wave step by

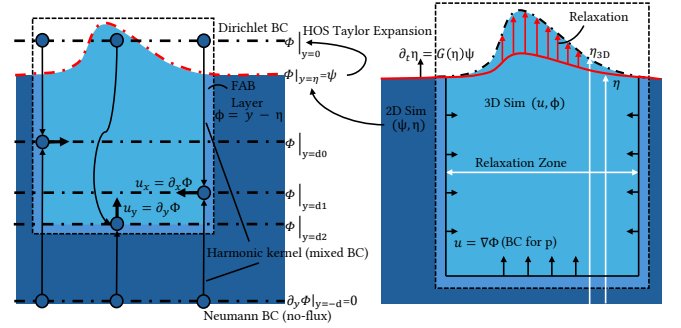


Fig. 4. Two-way coupling between the 2D wave solver and the 3D NS solver. **Left, 2D-to-3D (Section 4):**  $(\eta, \psi)$  is lifted to  $y = 0$  via the Craig–Sulem (HOS) Taylor expansion and propagated to depth layers by finite-depth harmonic kernels under Dirichlet top and Neumann no-flux bottom  $\partial_y \Phi|_{y=-d} = 0$ . The FAB ring on the box faces takes its level-set BC from the 2D height field via  $\phi = y - \eta$  and its velocity BC from  $\mathbf{u} = \nabla \Phi$ . **Right, 3D-to-2D (Section 5):** an exponential-decay source on the box footprint relaxes the wave-side  $\eta$  toward the 3D surface  $\eta_{3D}$  implicitly inside the relaxation zone, while outside the box the Zakharov equation  $\partial_t \eta = G(\eta)\psi$  evolves the ocean freely.

operator splitting (Step 8 of Alg. 2). Each half is therefore integrated exactly over  $\Delta t$ , and the time-step constraint never tightens with  $\sigma_{\text{max}}$ . After each step we apply the spectral state filter of Craig and Sulem [1993],

$$A(\mathbf{k}) = \prod_{\alpha \in \{x, z\}} \frac{1}{8} \left[ 5 + 4 \cos(\pi |k_\alpha| / k_{\text{Nyq}}^\alpha) - \cos(2\pi |k_\alpha| / k_{\text{Nyq}}^\alpha) \right], \quad (17)$$

to  $(\eta, \psi)$ . The filter is unity at low wavenumbers, smoothly suppresses the highest modes, and removes residual high-frequency noise without affecting the resolved spectrum.